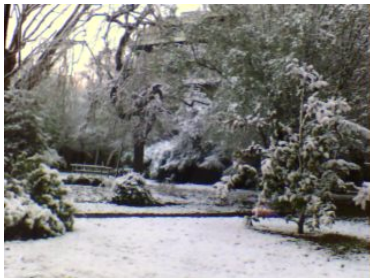


March 2008

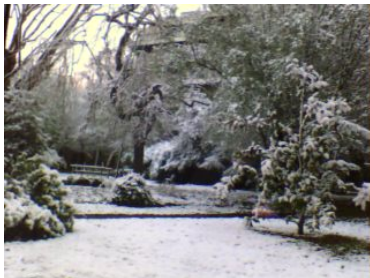
Pre high-frequency world

All seems to be smooth and regular, like a landscape covered by snow.



Post high-frequency world

Now we discover some singularities: the world is not as linear as we thought.



What fair value?

When the *fair value* of a firm changes, the price does not immediately “jumps” from one value to another one: the process of price formation takes place.

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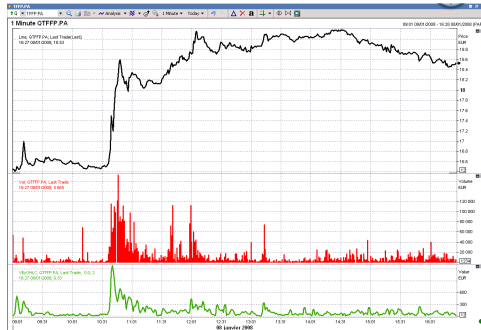
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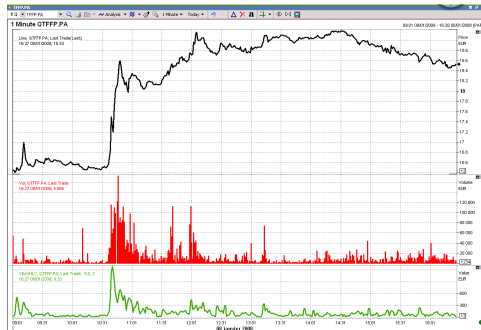
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Here we will not focus on the fair value estimation process, but on **how to deal with the price formation process.**

That is real

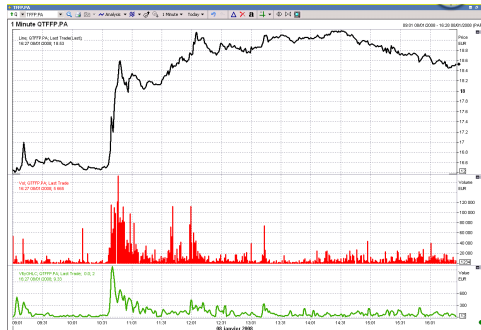


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This is the effect of the announcement by president Sarkozy that public TV channels will no more be allowed to sell advertising...

That is real



This is the effect of the announcement by president Sarkozy that public TV channels will no more be allowed to sell advertising...

The value of the main french private Channels jumped immediately.

CA CHEUVREUX

Effets of such a change

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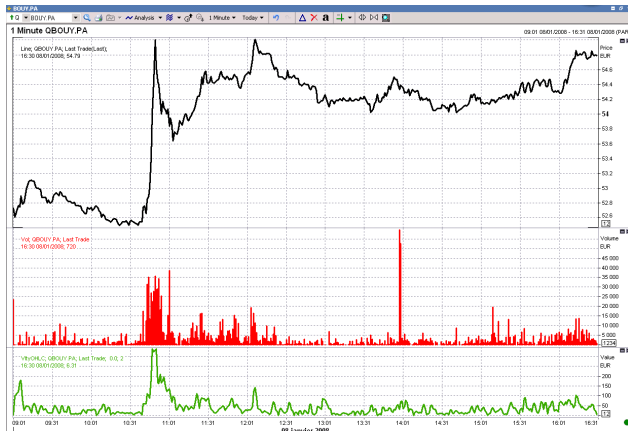
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This vision is the Capital Asset Pricing Model (CAPM) one: a worldwide market driven by **few systematic factors**. In this framework some temporary explorations are conducted by isolated stocks or sectors (**specifics explorations**).

Extra day arbitrageurs focus on those aspects. At intra day scale, we try to understand and detect when and how auction mechanisms of market places are able to absorb such trajectories.

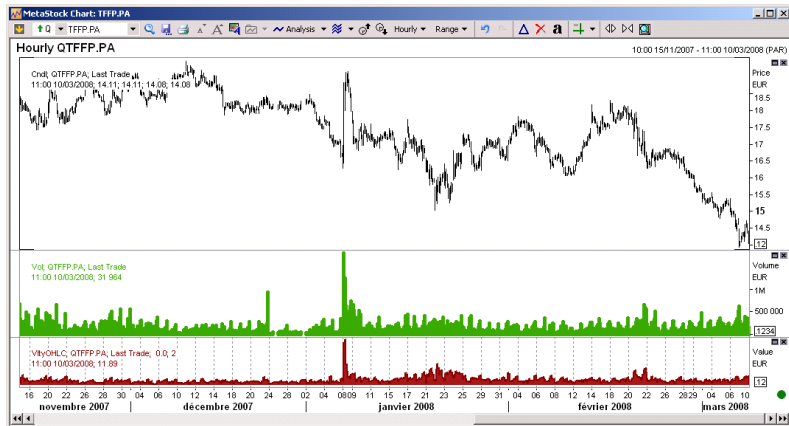
Propagation - correlations



Propagation - zooming out: 30 minutes



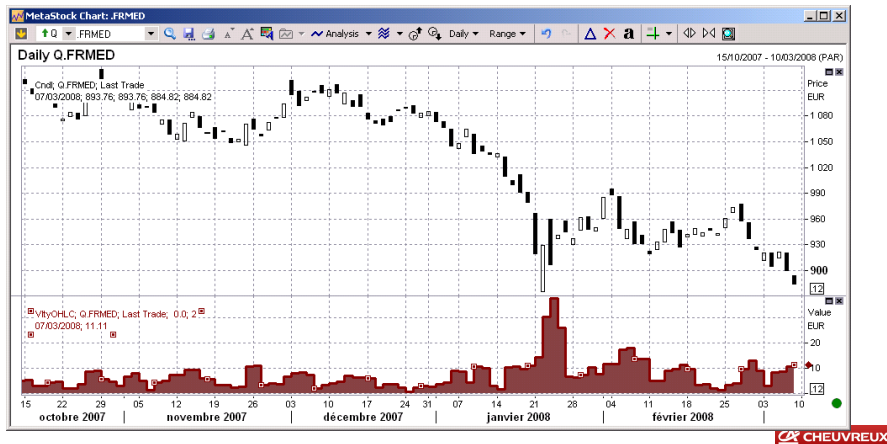
Propagation - zooming out: 1 hour



Propagation - zooming out: daily



Propagation - sector: daily media



Contenu

Practical aspects of auction markets

Bid-ask spread, limit order books

Intra day nomograms

The two main enemies of intraday trading

Alternate sources of liquidity

Quantitative optimization of high freq trading

High freq trading in equations

In the heart of darkness: market impact models

From simple to sophisticated optimization

Two main evolutions of stock trading

Non parametric approaches

Misc

The principles of an auction market

Some agents wants to buy, others want to sell. Three main roles:

- **Informed traders:** they are convinced to have a clear idea of the fair value of the equity part of the firm
- **Noise traders:** they need to trade for another reason (their utility function is not directly related to the price of the equity)

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Each agent place orders at given prices for given quantities, when they match: a transaction occurs

Bids and asks: from orders to LOB

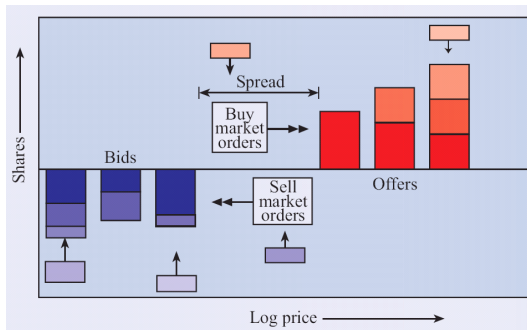
Quote: 0#UNBP.PA									
0#UNBP.PA		UNTBAIL	RODAMCO	FR0000124711	UL	PAR EUR	10APR08	12:07	
XD /	Last	TrdVol	Pct.Chng	Order	Bid Size	Bid 12:07 Ask	Ask Size	Order	
↑	162.61	21	-1.76 %	1	304	162.38 / 162.61	68	1	
	162.42	Net.Chng	-2.92	1	304	162.38 / 162.61	68	1	
	162.51	Volume	101875	1	100	162.37 / 162.62	372	2	
	162.51			1	100	162.35 / 162.82	358	1	
	162.51	Turnover	News	1	357	162.19 / 162.86	188	2	
				1	100	162.18 / 162.87	325	3	
VWAP 163.2907 16635239									
Moves 1145									
Up/LoLimit:174.22 / 157.64 FMP 400000 162.15 / 163.07 400000 FMP									
Close	: 167.23	09APR08	Yr.High	: 236.90	10APR07	Dividend	: 7.00		
Open	: 165.93		Yr.Low	: 136.51	22JAN08	Div.Dat	: 15APR08		
High	: 165.94		Yr.Perform.	: +11.55 %		Ex.Date	: 10APR08		
Low	: 161.55		Pr.Yr.Perf.	: -17.80 %		Yield	: 4.186 %		
Adj.Cls	: 165.53		PreStl	: 167.23		Earnings	: 15.00		
Adj.Fact	: 0.9898	10APR08	Report	: 172.190	25OCT07	P.E	: 11.15		
NrShares	: 81772493		% Liq	: -5.564 %					
Cap(b)	: 13.53580								
Options	: 0#UNBP.A*.p	Background	: UNBP.PAB2	LOC	: FR0000124711				
SEGM A	: 5.00	Sector	: UNBP.PAB2	SRD	: FR0000124711				
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Price is a function of quantity!

From orders to LOB



Statistical theory of the continuous double auction - Eric Smith , J
Doyne Farmer, Laszlo Gillemot and Supriya Krishnamurthy - Santa
Fe Institute - September 2003

- ▶ Wait to be executed (not executed for sure)
- ▶ Execution price is better than the last quoted one
- ▶ (sometimes) pays less fees

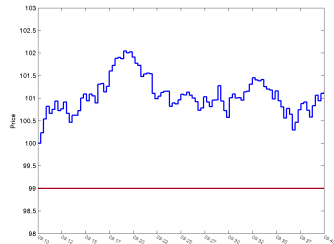
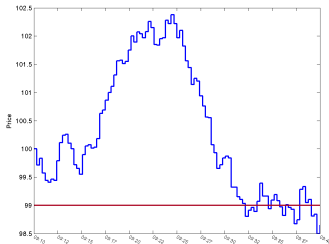
- ▶ Executed for sure
- ▶ Pays around half a bid-ask spread more than the last quoted price
- ▶ Pays *Price Impact* for large quantities

A non-singular model is obtained by homogenizing B in the style of K

The free option myth

If I decide to place a *Buy* order (one share) at a price of $S_0 - \delta S$ as a liquidity provider during ΔT seconds, and then to buy at $S_{\Delta T}$ as a consumer (if I have not been executed): *it's as simple as a barrier option*.

A very special one: k.o. option with barrier level $B =$ the strike K .



Should I stay or should I go?

Let q_T be the probability to cross the barrier between 0 and T (τ_B is the first time of crossing), bayes decomposition gives us:

$$E(S) = E(S|\tau_B < T) \underbrace{P(\tau_B < T)}_{q_T} + E(S|\tau_B \geq T)P(\tau_B \geq T)$$

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$\Rightarrow$ All the complexity of intra day trading comes from market microstructure and price / volume dependencies.

My first continuous auctions formula

Let $f_{\text{LOB}}^-(s)$ be the volume available in the LOB by sellers (ask prices) at price s (resp. $f_{\text{LOB}}^+(s)$ for bids quantities). When I want to buy a quantity V of shares of this instrument, I will generate a **Market Impact** of S_{MI} implicitly defined by the equation (S_0 is the last quoted price):

$$(1) \quad V = \int_{s=S_0}^{S_{MI}} f_{\text{LOB}}^-(s) ds$$

and my **Volume Weighted Average Price (VWAP)** will be:

$$(2) \quad \text{VWAP} = \frac{1}{V} \int_{s=S_0}^{S_{MI}} s \cdot f_{\text{LOB}}^-(s) ds$$

One fundamental order book equation

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"The big secret...Quantitative finance is one of the easiest branches of mathematics"

Paul Wilmott on his blog, April 2008...

Fixing auctions

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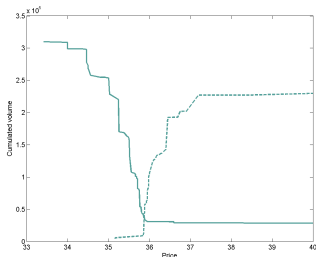
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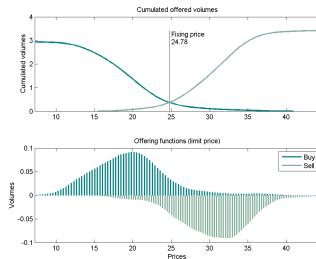


- ▶ during the *pre fixing* phase, orders are not crossed to generate transactions
- ▶ a “*theoretical price*” is published
- ▶ at the end (5 minutes later): the fixed price is the natural Pareto equilibrium

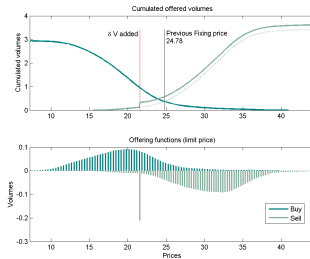
Fixing price

The fixing price S^* is such that

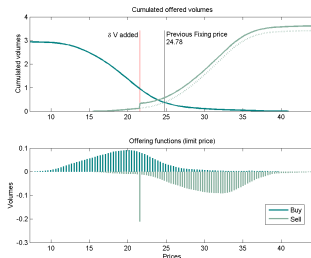
$$(4) \quad F_{\text{LOB}}^+(S^*) = F_{\text{LOB}}^-(S^*)$$



If I put a quantity δV into the fixing orderbook at a price $\tilde{S}$ into F_{LOB}^- .

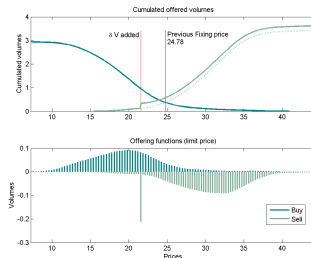


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$$(5) \quad F_{\text{LOB}}^+(\bar{S}) = F_{\text{LOB}}^-(\bar{S}) + \delta V \cdot \mathbf{1}_{\tilde{S} < S^*}$$

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It is a matter of slopes. The effect of adding volume far from the previous equilibrium price S^* is mainly related to **local properties** of the orderbook around S^* .

From price impact to market impact

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- ▶ **Dynamical market depth.** The more the volatility of the price diffusion is high, the more the LOB is structurally empty: M.I. increases with σ .
- ▶ **Usual quantities:** the more my volume v is large compared to the “usual” traded volume $V_{\Delta T}$ (during this time interval ΔT), the more my M.I. increases.

Qualitative Market Impact model

Basic Market Impact model

$$(7) \quad \text{MI}(v) = \psi + F\left(\sigma, \frac{v}{V_{\Delta T}}\right)$$

where $F(\cdot, \cdot)$ increases with respect to its two inputs.

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Usually σ is on the risk side of the equation, because market impact is subtracted to your return, we already see that in a quantitative trading optimisation framework σ is (at least) on the return side...

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What do we need?

As seen is equation (7), we need to capture σ , usual volumes, and half bid-ask spread.

What means “*capture statistical invariants*”

- ▶ No risk-neutral measure available,

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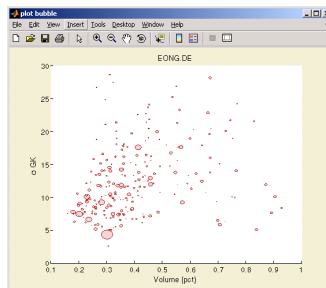
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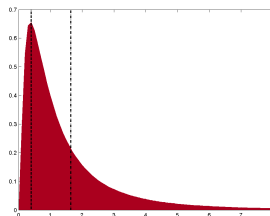
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Volumes: do not use "average" on log normal distributions



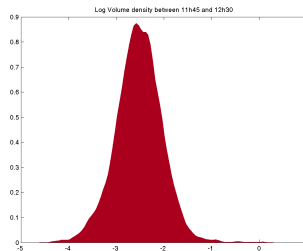
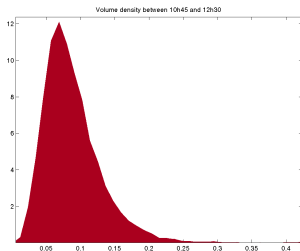
Density of a log normal distribution:

$$(8) \quad f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp - \frac{(\ln(x) - \mu)^2}{2\sigma^2}$$

So (the median is not equal to the expectation):

$$M = E(X) = \exp \mu + \frac{\sigma^2}{2}$$

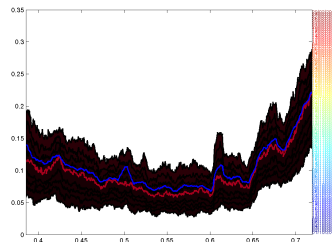
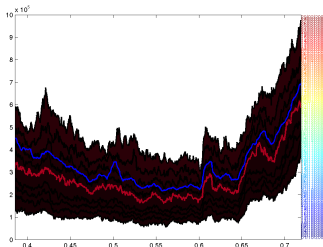
Five months of FTE.PA, between 11h45 and 12h30



Log normality: what do we really need

It is easy to see that $P(X > M) \simeq 30\%$. Do we want the mean value, or the median?

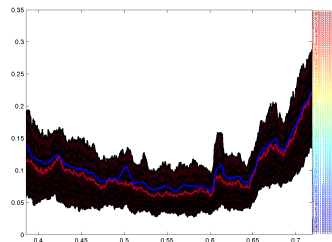
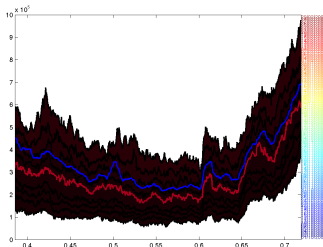
Five months of FTE.PA (05/07 to 03/08) capital vs proportions



Log normality: what do we really need

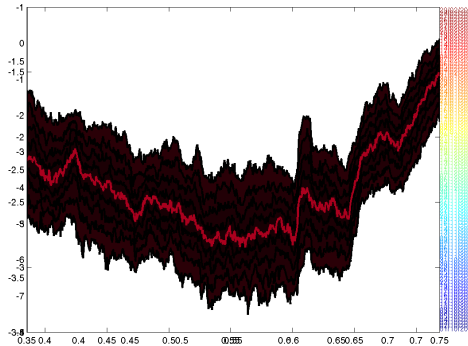
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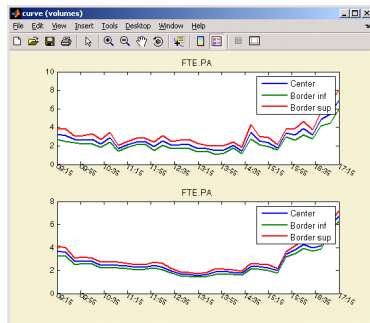
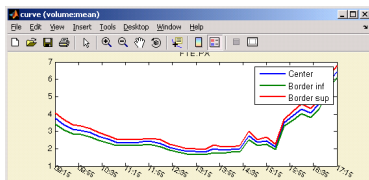


If you really want the mean, use robust mean (expectation using median and truncated variance) instead of average!

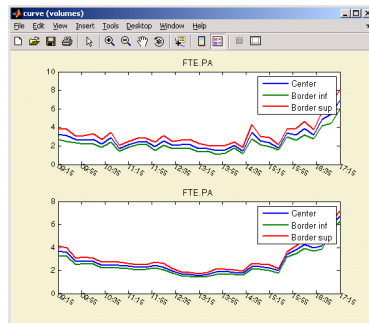
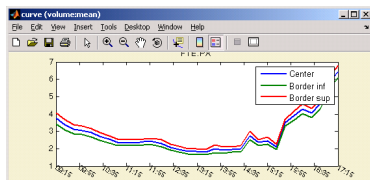
Log of proportions



Do not mix heterogeneous effects

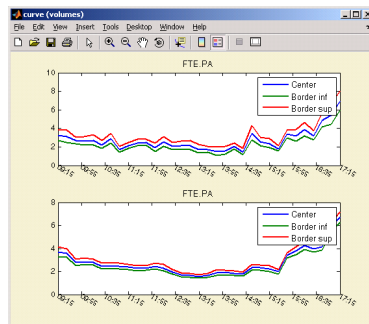
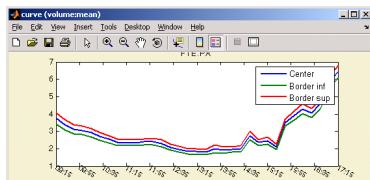


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A large part of the variance comes from mixing Fridays with other days.

Do not mix heterogeneous effects



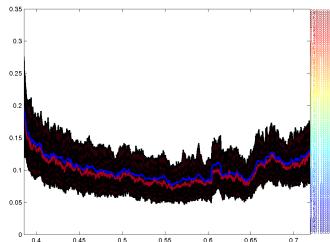
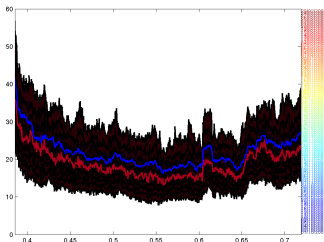
A large part of the variance comes from mixing Fridays with other days.

You can use auto correlations to obtain more robust estimators  CHEUVREUX

Volatility : do only add squared volatilities !

We all know that $\mathcal{N}(0, \sigma_1) + \mathcal{N}(0, \sigma_2) \sim \mathcal{N}(0, \sqrt{\sigma_1^2 + \sigma_2^2})$.

Five months of FTE.PA (05/07 to 03/08) σ_{GK} vs proportions



Volatility are not “as lognormal” as volumes are. Mainly, they are “far less i.i.d.” ...

Intra day volatility modelling

Because of the Markovian aspect of σ intra day, we would like to model it through a diffusion process.

A natural candidate is the C.I.R. model:

$$(9) \quad d(\sigma^2) = \gamma \cdot (m - \sigma_t^2) dt + \kappa \sigma_t dW_t$$

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The point is that m and κ should be time dependent here...

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Our prices are discretized (rounded?) on a price grid

A lot of interesting papers have been written on this subject:

Jacod, Delattre, At-Sahalia, Zhang, Mykland, Shepard, Rosenbaum

A “*simple*” model:

$$(10) \quad X_{n+1} = X_n + \sigma \sqrt{\delta t} \xi_n, \quad S_n = X_n + \varepsilon$$

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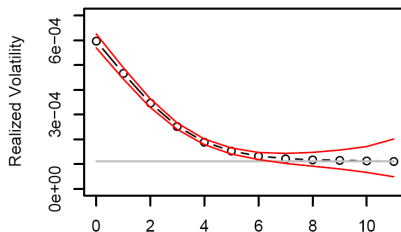
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thanks to Mathieu Rosenbaum for this plot

The simplest intra day volatility estimator

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- ▶ More robust to micro structure effects

Bid-ask spread : when microstructure limits the use of diffusion models

If only one microstructure effect should be kept, it would be the Bid-Ask spread:

- ▶ Sell price $\neq$ Buy price

Bid-ask spread : when microstructure limits the use of diffusion models

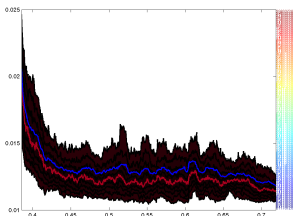
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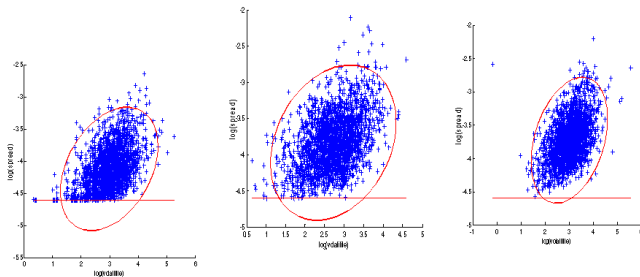
If only one microstructure effect should be kept, it would be the Bid-Ask spread:

- ▶ Sell price $\neq$ Buy price
- ▶ Volume has an influence on the price
- ▶ Volatility estimations are not so simple



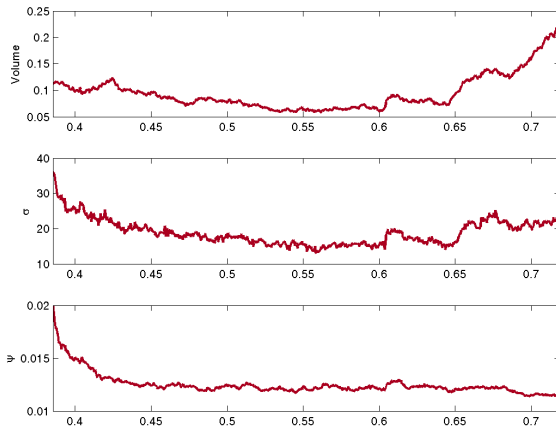
What about real statistics?

“Spread effect” on three stocks





Intra day nomograms



Give me two of the three curves, I will give you the missing one...

Market impact (volume driven) demands to trade slowly

As suggested by equation (7), a good way to reduce your Market Impact is to trade on a larger time interval ΔT .

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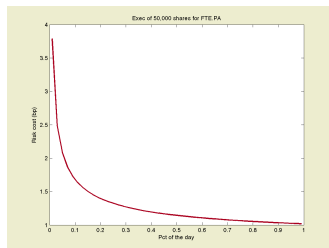
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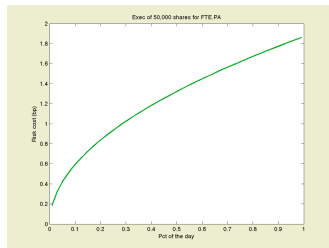
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Continuous auctions mechanisms

Two main behaviours:

- ▶ Liquidity provider: provide orders to the orderbook (a quantity at a given price for a defined side), my orders are consolidated inside the *“limit order book”*.
- ▶ Liquidity consumer: give orders (*at best, price limited*, or more complex) which consume one or more orders of the LOB.

Each market place (*Dark Pools, Multilateral Trading Facilities*, etc) has this set of specific orders, like *iceberg, pegged* or *totally hidden*).

Multilateral trading facilities (MTF)

Chi-X, turquoise, etc... are here or are coming

Quote: BNPPpa.CHI

BNPPpa.CHI	Type:	BNP	PARIBAS	FR0000131104	INS/102	EUR	09APR08	11:07
Last	Net.Chng	50	200	68.4950x68.5300	400			
68.5000	-0.9050	50	200	68.4900x68.5300	400			
68.5000	Pct.Chng	Volume	560	68.4850x68.5450	198			
68.5000	-1.3 %	60098	200	68.4800x68.5500	200			
68.5000	Buy-Sell	ID	Value	880	68.4700x68.5550	500		
68.5000	- / -	4131007.25	380	68.4650x68.5600	200			
	Moves			68.4550x68.5650	340			
5231	326	15	400	68.4350x68.5700	200			
Open	High	52.Wk.Hi:	1011	68.0800x68.5750	880			
68.8700	69.1700		1000	68.0500x68.5800	380			
CLs:08APR08	Low	52.Wk.Lo:						
69.4050	68.2900							

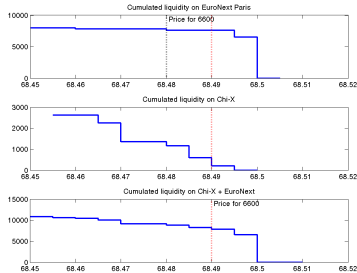
Quote: 0#BNPP.PA

0#BNPP.PA	Type:	BNP	PARIBAS	FR0000131104	INS/102	EUR	09APR08	10:08
Last	TrdVol	Pct.Chng	Order	Bld Size	Bld	Ask	Ask Size	Order
68.5000	100	-1.69 %	11	6524	68.5000 / 68.5450	359	2	
68.5450	Net.Chng	-1.18	2	1074	68.5000 / 68.5450	359	2	
68.5450	Volume	744392	1	200	68.4950 / 68.5500	396	3	
68.4950	Turnover	News	1	200	68.4800 / 68.5550	344	1	
68.4900	51157744	09:03	1	202	68.4600 / 68.5650	200	1	
YUAP	51157744				68.4500 / 68.5700	200	1	
Moves	2234							
Up/LoLimit:	72.3650 / 65.4750							

More order types: totally hidden, pegged, etc...

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Illuminating the new dark influence on trading and U.S. market structure, Carl Carrie, **JOT** Winter 2008

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What does mean VWAP of an order in such a context?

Equations

I have a volume V^* to buy from 0 to T ; I will assume a regular time grid of width δt (n from 1 to $N = \lceil T/\delta t \rceil$).

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So my total price is:

$$(13) \quad W = \sum_{n=1}^N v_n \cdot \underbrace{(S_n + \eta_n(v_n))}_{\tilde{S}_n(v_n)}$$

The classical rewriting

$$x_n = \sum_n^N v_n, \quad x_1 = V, \quad x_{N+1} = 0$$

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$$(14) \quad W = V \cdot S_0 + \sum_{n=1}^N x_n \sigma_n \sqrt{\delta t} \xi_n + \sum_{n=1}^N v_n \cdot \eta_n(v_n) + V \cdot \alpha T$$

Cost minimization

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Trade regularly, **follow the market**...

Risk minimization

Risk minimization

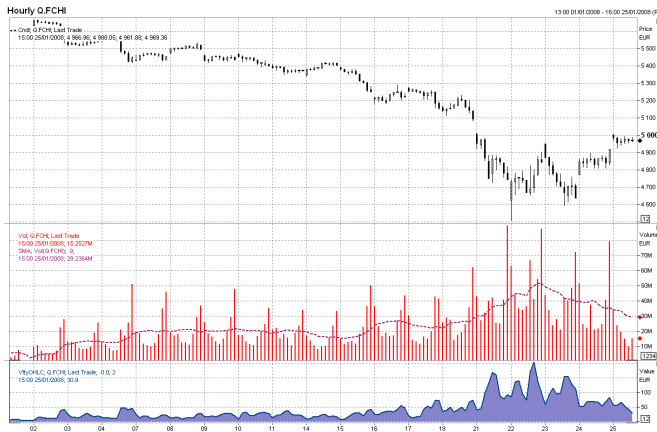
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Risk minimization

$$\mathbf{V}(W|\sigma, V, \alpha) = \sum_{n=1}^N x_n^2 \sigma_n^2 \delta t$$

Trade all your quantity during the first time interval!

A market impact effect?



Market Impact main model

With ψ the half spread, σ the volatility (in currency), and V a constant homogeneous with a quantity of shares.

The most common model is:

$$(15) \quad f_m(v) = \psi + \kappa \sigma \cdot \left(\frac{v}{V} \right)^\gamma$$

This M.I. model put inside equation (3) gives:

$$f_{\text{LOB}}^{-} \left(S_0 + \psi + \kappa \sigma \left(\frac{v}{V} \right)^{\gamma} \right) \cdot \kappa \sigma \gamma \frac{v^{\gamma-1}}{V^{\gamma}} = 1$$

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using $h = \psi + \kappa \sigma (v/V)^{\gamma}$; i.e. $v = V ((h - \psi)/\kappa \sigma)^{1/\gamma}$
we obtain:

$$(16) \quad f_c(S_0 + h) = \frac{V}{\kappa \sigma \gamma \left(\frac{h - \psi}{\kappa \sigma} \right)^{\frac{\gamma-1}{\gamma}}}$$

Equation (16) implies that $\gamma \leq 1$, besides:

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A square root market impact

A square root market impact implies a linear LOB.

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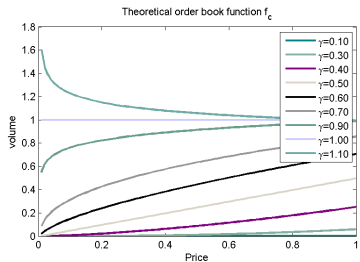
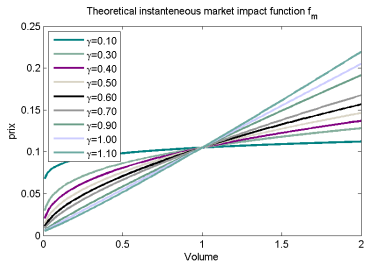
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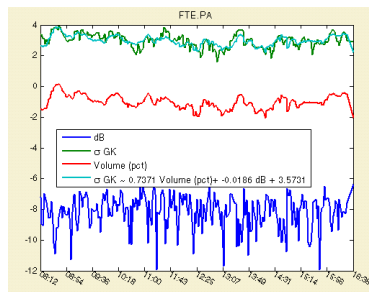
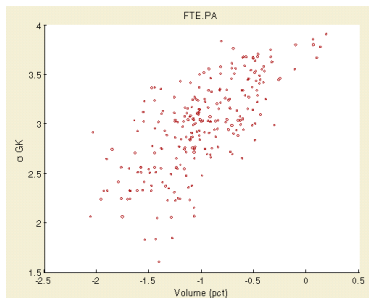
With $\gamma = 1/2$, we obtain

$$f_c(S_0 + h) = 2(h - \psi)V$$

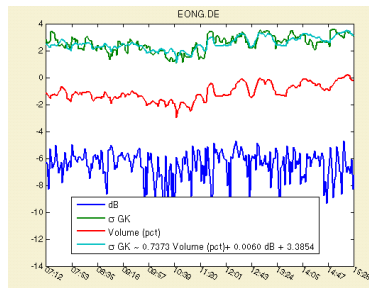
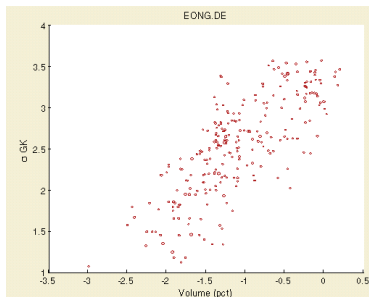
In the heart of darkness: market impact models



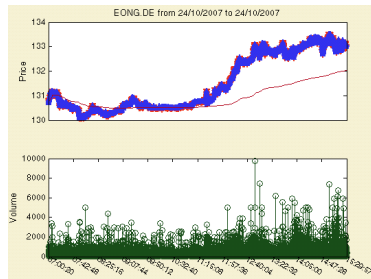
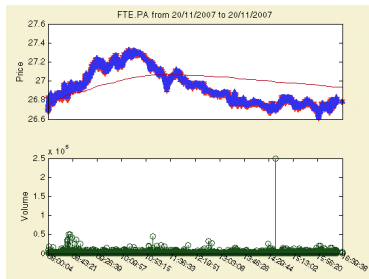
Some evidences: FTE.PA 20/11/2007



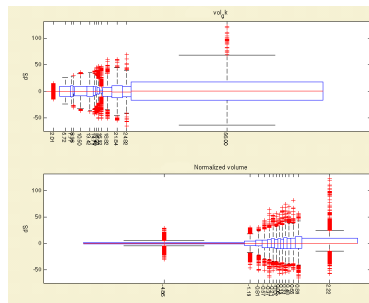
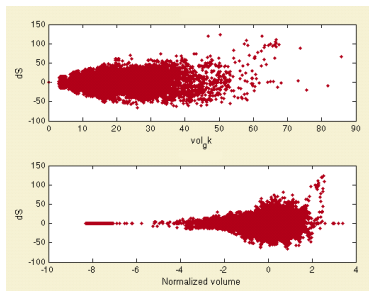
Some evidences: EONG.DE 24/10/2007



FTE and EOND trading data

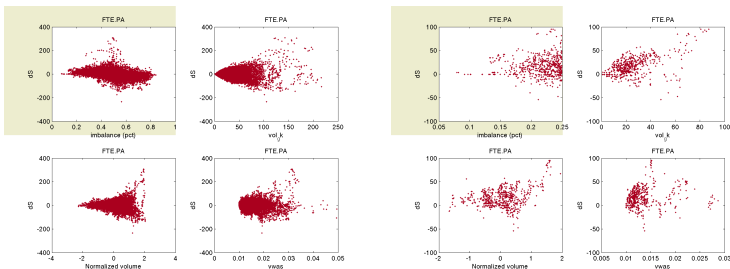


6 months 30 sec by 30 sec data for one stock



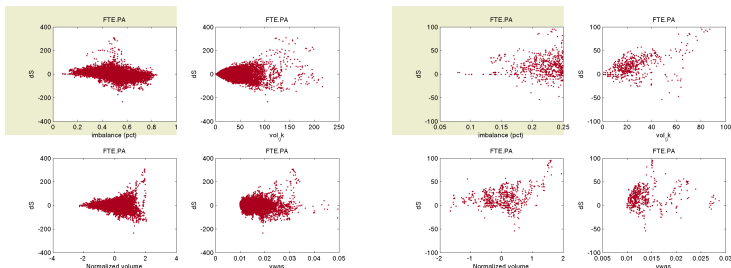
Standard pitfalls

- Do not use naively absolute value on δS (have a look at imbalance)



Standard pitfalls

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- Really think about choosing the right ΔT

Simplest case: constant volatility and volumes

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We will minimise (read Almgren, Chriss)

$$(17) \quad \mathcal{C}_1(v) = E(W|\sigma, V, \alpha) + \lambda \cdot \mathbf{V}(W|\sigma, V, \alpha)$$

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λ is a **risk aversion** parameter.

Solution for constant volatility and volumes

No need for Lagrangian multipliers (thanks to the $v \rightarrow x$ change of variable)

(18)

$$x_{n+1} = \left(1 + \frac{\sigma_{n-1}}{\sigma_n} \cdot \frac{V_n}{V_{n-1}} + \lambda \sigma_n V_n \delta t \right) x_n - \frac{\sigma_{n-1}}{\sigma_n} \cdot \frac{V_n}{V_{n-1}} \cdot x_{n-1}$$

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And when $\sigma_n \equiv \sigma$, $V_n \equiv \bar{V}$, we obtain this linear equation:

$$(19) \quad x_{n+1} = (2 + \lambda \sigma \bar{V} \delta t) \cdot x_n - x_{n-1}$$

its “initial conditions” are $x_1 = V$, $x_{N+1} = 0$

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What about $E(1/V_n)$? If $V_n \perp V_k$ when $n \neq k$

$$\mathbf{V}(W|\sigma, \alpha = 0) = \sum_{n=1}^N x_n^2 \sigma_n^2 \delta t + \eta^2 \sum_{n=1}^N (x_n - x_{n+1})^4 \sigma_n^2 \cdot \mathbf{V}\left(\frac{1}{V_n}\right)$$

else...

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$$\mathbf{V}(W|\alpha = 0) = \sum_{n=1}^N x_n^2 \underbrace{E(\sigma_n^2)}_{\int_n^{(n+1)\delta t} \sigma_t^2 dt} \delta t + \eta^2 \sum_{n=1}^N (x_n - x_{n+1})^4 \cdot \mathbf{V}\left(\frac{\sigma_n}{V_n}\right)$$

Towards stochastic control

Let use a continuous framework (read Almgren and Lorenz); v rate of buying:

$$(22) \quad \begin{cases} dx &= -v dt & \text{remaining shares} \\ dy &= (s + \eta v) v dt & \text{spent money} \\ ds &= \alpha dt + \sigma dW & \text{stock price} \end{cases}$$

We want to minimize $E(y(T))$ so that $x(T) = 0$, let u be the *value function* of the solution (min from t).

Hamilton-Jacobi-Bellman:

$$(23) \quad \partial_t u + \min_v \{-v \partial_x u + v(s + \eta v) \partial_y u + \alpha \partial_s u + \frac{1}{2} \sigma^2 \partial_s^2 u\} = 0$$

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$$v^* = \frac{s \partial_y u - \partial_x u}{2 \eta \partial_y u}$$

replace v^* into equation (23), and solve in u . Then replace in (24)... done.

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"The big secret... Quantitative finance is one of the easiest branches of mathematics"

Paul Wilmott on his blog, the 1st of April 2008...

Market making

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- ▶ At time $n\delta t$ the half spread is ψ , the mid price is $\tilde{S}$, the LOB is known, you decide to offer quantities at prices $\tilde{S} - \delta s^-$ and $\tilde{S} - \delta s^+$
- ▶ You “*control*” your trading not by quantities but by limit prices $\delta s^\pm$
- ▶ The usual v_n are now **Random variables** conditioned by the distance between your offer and the market LOB
- ▶ More probability theory... less closed form formula...



Portfolio trading

Equation (12) is now

$$(25) \quad S_{n+1}^{(k)} = S_n^{(k)} + \alpha^{(k)} \delta t + \sigma_{n+1}^{(k)} \sqrt{\delta t} \xi_{n+1}^{(k)}$$

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- ▶ Similarly to volatility, how can we estimate tick-by-tick correlations? What about synchronisation of high freq datasets ? (see Zhang, Hayashi, Yoshida)
- ▶ But Central Limit Theorem should help...

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Anyway multi stocks view is very useful: Market Impact can be seen as a relative move between two usually highly correlated stocks.

Implicit market impact models

- ▶ Based on non parametric statistics
- ▶ Kernel / local methods (read Devroye, Lugosi)

Let $K(x)$ a kernel function (symmetric positive measure with its max at zero), we know that:

$$(26) \quad f^*(x) = \frac{1}{N \cdot h(N)} \sum_{n=1}^N K\left(\frac{x - X_n}{h(N)}\right)$$

is a good (non parametric) estimate of the density of the law of $(X_n)_{n \in \{1, \dots, N\}}$; in **local regressions**, such kernels are used to weight observations

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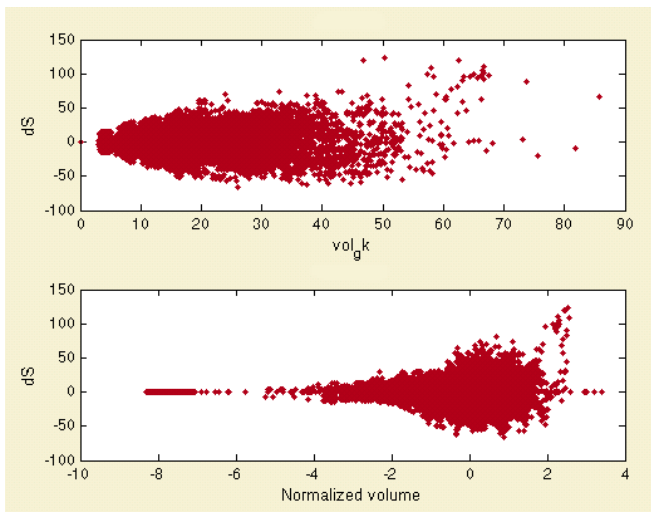
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- ▶ It is like “*smoothly*” look around similar market conditions in the past, going back to an estimation framework:

$$\text{MI}(v) = E(\delta S | v, \text{market conditions})$$

Direct use of such models



Interpretation of risk aversion

- ▶ Equation (17) involves a risk aversion parameter λ .
- ▶ What about the risk aversion parameter of asset allocation theory?

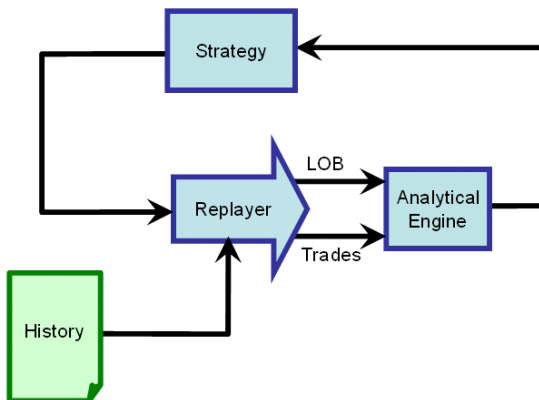
Interpretation of risk aversion

- ▶ Equation (17) involves a risk aversion parameter λ .
- ▶ What about the risk aversion parameter of asset allocation theory?
- ▶ For portfolio trading, this is a real question
- ▶ Will you use the same risk aversion parameter in you Markowitz / Black-Litterman allocation and your trading strategy?

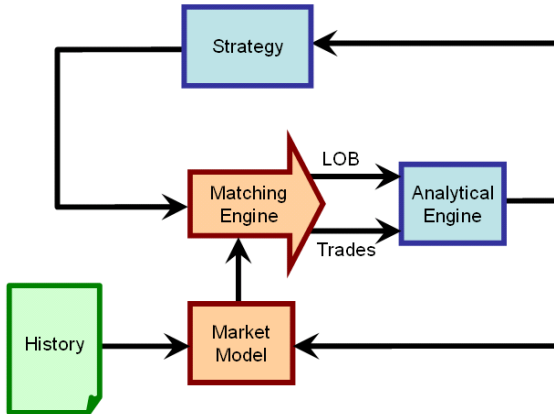
(read Engle, Ferstenberg)

Focus on back tests

“Naive” backtest framework



“Really” backtest framework



Conclusion (1/2)

- ▶ Intra day trading is affected by
 - ▶ Intra day historical invariance (volumes, volatility, spread nomograms)
 - ▶ Dependencies / correlations between those variables
 - ▶ Price discretisation (bid-ask bounce can be greater than volatility)

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 - ▶ Intra day historical invariance (volumes, volatility, spread nomograms)
 - ▶ Dependencies / correlations between those variables
 - ▶ Price discretisation (bid-ask bounce can be greater than volatility)
- ▶ We have seen how such effects can be captured using (not straight forward) statistics
 - ▶ Take into account the tick size (volatility and spread)
 - ▶ Identify trading contexts to build dedicated intra day nomograms (Friday effect)
 - ▶ Do not forget to look at auto correlations

Conclusion (1/2)

- ▶ There is a quantitative framework to deal with all those effects
 - ▶ Optimise to find the balance between two main effects: Market Impact demands to trade slowly and Market Risk demands to trade fast
 - ▶ Be sure to have accurate estimates for the parameters of your equations

Conclusion

- ▶ Intra day trading is affected by micro structure effects
- ▶ We have seen how such effects can be captured using (not straight forward) statistics
- ▶ There is a quantitative framework to deal with all those effects